# A GRADED COHOMOLOGICAL BOUNDARY-MODULE CRITERION FOR UNIFORM MITTAG-LEFFLER STABILIZATION

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*Editorial note.* The surrounding source explicitly works with inverse systems of A/I^n-modules. The proof uses this when lift-then-multiply is independent of the lift and when a correction vanishes at level n-c.

*Editorial note.* The degree-zero piece is H^p(X,F_1), and the finite number of initial stages does not affect the Mittag-Leffler conclusion.

*Editorial scope.* One original lemma with full proof; context makes explicit that the system consists of A/I^n-module sheaves. No I-adic topology claim is added.

*Allowed standard prerequisites (editorial).*

- Sheaf cohomology long exact sequence and naturality. A short exact sequence induces connecting maps natural in a commutative diagram.
- Noetherian graded modules. The ascending chain condition implies a graded submodule has finitely many homogeneous generators.

Let $A$ be a ring and let $I \subset A$ be an ideal. We prove some results on inverse systems of sheaves of $A/I^n$-modules.

**Lemma 0.1.** *Let $I$ be an ideal of a ring $A$. Let $X$ be a topological space. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

*be an inverse system of sheaves of $A$-modules on $X$ such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Assume*

$$\bigoplus\nolimits_{n\geq 0} H^{p+1}(X, I^n\mathcal{F}_{n+1})$$

*satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module. Then the inverse system $M_n = H^p(X, \mathcal{F}_n)$ satisfies the Mittag-Leffler condition[1].*

*Proof.* Set $N_n = H^{p+1}(X, I^n\mathcal{F}_{n+1})$ and let $\delta_n : M_n \to N_n$ be the boundary map on cohomology coming from the short exact sequence $0 \to I^n\mathcal{F}_{n+1} \to \mathcal{F}_{n+1} \to \mathcal{F}_n \to 0$. Then $\bigoplus \mathrm{Im}(\delta_n) \subset \bigoplus N_n$ is a graded submodule. Namely, if $s \in M_n$ and $f \in I^m$, then we have a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I^n\mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \\ 0 & \longrightarrow & I^{n+m}\mathcal{F}_{n+m+1} & \longrightarrow & \mathcal{F}_{n+m+1} & \longrightarrow & \mathcal{F}_{n+m} & \longrightarrow & 0 \end{array}$$

The middle vertical map is given by lifting a local section of $\mathcal{F}_{n+1}$ to a section of $\mathcal{F}_{n+m+1}$ and then multiplying by $f$; similarly for the other vertical arrows. We conclude that $\delta_{n+m}(fs) = f\delta_n(s)$. By assumption we can find $s_j \in M_{n_j}$, $j = 1, \ldots, N$ such that $\delta_{n_j}(s_j)$ generate $\bigoplus \mathrm{Im}(\delta_n)$ as a graded module. Let $n > c = \max(n_j)$. Let $s \in M_n$. Then we can find $f_j \in I^{n-n_j}$ such that $\delta_n(s) = \sum f_j\delta_{n_j}(s_j)$.

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[1] In fact, there exists a $c \geq 0$ such that $\mathrm{Im}(M_n \to M_{n-c})$ is the stable image for all $n \geq c$.

We conclude that $\delta(s - \sum f_j s_j) = 0$, i.e., we can find $s' \in M_{n+1}$ mapping to $s - \sum f_j s_j$ in $M_n$. It follows that

$$\mathrm{Im}(M_{n+1} \to M_{n-c}) = \mathrm{Im}(M_n \to M_{n-c})$$

Namely, the elements $f_j s_j$ map to zero in $M_{n-c}$. This proves the lemma. $\square$